

Notes: Hendel and Nevo (Econometrica, 2006)

Measuring the Implications of Sales and Consumer Inventory Behavior

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1 Big picture

1.1 Main question

The paper asks how much standard static demand estimation is biased when consumers can store products and time their purchases around temporary price reductions. For storable goods, a sale can generate a large spike in purchases even if long-run consumption changes little. Thus the short-run quantity response to a sale need not equal the long-run response to a permanent price change.

1.2 Core answer

The authors estimate a dynamic model in which households choose whether to buy, what brand-size combination to buy, and how much to consume while carrying inventory over time. The estimates imply that ignoring inventory behavior materially changes demand elasticities. Relative to the dynamic model, a static model:

- overstates own-price elasticities by roughly 30 percent;
- understates cross-price elasticities to other products, in some cases by a factor close to five;
- overstates substitution to the outside option or no-purchase option by more than 200 percent.

These differences matter for price-cost margins, merger simulation, market definition, and any policy exercise that depends on long-run substitution patterns.

1.3 Why this paper matters

Demand estimates are often used as inputs into industrial organization policy and firm-behavior models. If static demand estimation confuses stockpiling with consumption demand, then it exaggerates the own-price response to a temporary sale and mismeasures substitution across products. This paper provides a structural way to separate timing responses from long-run consumption and brand-substitution responses.

2 Incremental contribution of the paper

2.1 Relative to static demand estimation

The paper's first contribution is to show that high-frequency scanner data cannot simply be treated as repeated static choice data when goods are storable. Temporary sales induce intertemporal substitution. Consumers buy earlier or buy more because the price is temporarily low, not necessarily because they consume much more in the long run.

2.2 Relative to previous stockpiling studies

Earlier marketing and economics studies documented facts consistent with stockpiling, such as purchase acceleration and longer duration to the next purchase after sales. Hendel and Nevo go further by estimating a structural model that can compute counterfactual long-run elasticities and compare them directly with static elasticities.

2.3 Relative to fully dynamic brand-inventory models

A fully dynamic model with inventories of every brand and a large vector of prices is computationally difficult. The paper’s main methodological innovation is a decomposition of the likelihood. Conditional on purchase size, the brand choice part is static. The remaining dynamic problem is a quantity-choice problem summarized by inclusive values. This makes a rich dynamic model computationally feasible with a large choice set.

2.4 Policy contribution

The paper connects consumer stockpiling to antitrust and merger analysis. Static elasticities suggest that consumers are more willing to stop buying and less willing to switch among rival products than they truly are in the long run. This can bias inferred margins downward and make merger effects look smaller than they actually are.

3 Data and setting

3.1 Scanner data

The paper uses Information Resources Inc. scanner data from June 1991 to June 1993. The data have two parts:

- **Store-level data:** weekly prices, quantities sold, and promotional variables for detailed brand-size products in nine supermarkets across two submarkets of a large midwestern city.
- **Household-level panel data:** store visits and household purchases in 24 product categories, including whether a household visited a supermarket and did not buy laundry detergent.

The empirical analysis focuses on liquid laundry detergent because the product is storable, widely sold on sale, and available in a manageable set of major sizes.

3.2 Industry features

Laundry detergent is a useful category because purchases are infrequent enough for stockpiling to matter. Liquid detergents account for about 70 percent of quantity sold. A few brands dominate the category, but households differ in their preferred brands. The median household buys detergent about every four weeks and purchases only a few brands over the two-year sample.

3.3 Sales and nonlinear pricing

The product category has substantial temporary price reductions. In the sample, about 43 percent of liquid detergent volume and 36 percent of powder detergent volume is sold on sale, where a sale is defined as a price at least 5 percent below the modal product-store price. Larger containers are cheaper per ounce, so storage costs and nonlinear pricing interact: a household with low storage costs may prefer buying a large container or buying during a sale, while a high-storage-cost household may not.

3.4 Preliminary stockpiling evidence

The authors emphasize several reduced-form facts consistent with consumer inventory behavior:

- Aggregate quantity purchased during a sale is higher when more time has passed since the previous sale.
- Sale purchases occur after shorter time since the previous purchase, which means households buy even when inventory is relatively high.
- Sale purchases are larger and are followed by longer duration until the next purchase.
- Measures related to storage costs help explain heterogeneity in households' tendency to buy on sale.

These facts motivate the structural model.

4 Model

4.1 Household utility

Household h receives per-period utility from consumption of detergent and the outside good:

$$U(C_{ht}, v_{ht}; \theta_h) + \alpha_h m_{ht},$$

where C_{ht} is total detergent consumption, v_{ht} is a stochastic utility shock, m_{ht} is outside-good consumption, θ_h captures household preferences, and α_h is the marginal utility of income.

4.2 Choices

In each period a household chooses:

1. whether to purchase detergent;
2. which brand j to buy;
3. which size or quantity x to buy;
4. how much to consume, c_{ht} .

Unconsumed detergent is stored. The inventory law of motion is

$$i_{h,t+1} = i_{ht} + x_{ht} - c_{ht}.$$

The household trades off the benefit of buying at low current prices against the storage cost of carrying inventory.

4.3 Dynamic problem

For a household, the dynamic problem can be written schematically as

$$V(s_t) = \max_{c_t, d_{jxt}} \mathbb{E} \sum_{\tau=t}^{\infty} \delta^{\tau-t} \left[u(c_{\tau} + v_{\tau}; \theta) - C(i_{\tau+1}; \theta) + \sum_{j,x} d_{jx\tau} (\alpha p_{jx\tau} + \xi_{jx} + \beta a_{jx\tau} + \varepsilon_{jx\tau}) \right],$$

subject to nonnegative inventory and the inventory law of motion. The state includes inventory, current prices and promotions, utility shocks, and idiosyncratic product shocks.

4.4 Key assumptions

The tractability of the model comes from several assumptions:

- Utility shocks are independent over time and across consumers.
- Prices and advertising follow an exogenous first-order Markov process.
- Product-choice shocks are independently and identically distributed extreme value type I.
- Brand-specific differentiation enters at the purchase stage, so only total inventory matters rather than a vector of inventories by brand.

The last assumption is important: a household does not need to keep track of separate Tide, Wisk, or All inventory stocks in the dynamic state.

5 Econometric strategy

5.1 Main difficulty

The direct nested fixed-point algorithm would be computationally burdensome because the state includes inventories and many prices and promotional variables for all brand-size combinations. Inventory is also unobserved by the econometrician.

5.2 Handling unobserved inventory

For any trial parameter vector, the model implies optimal consumption. The authors generate an initial inventory distribution and then update inventory period by period using observed purchases and model-implied consumption. This makes the likelihood of a sequence of purchases computable even though actual inventory is not observed.

5.3 Three-step likelihood decomposition

The central computational contribution is a three-step estimator.

Step 1: static brand choice conditional on size. Conditional on the size purchased, the probability of choosing brand j does not depend on the dynamic inventory problem. It takes the logit form

$$\Pr(d_{jt} = 1 \mid x_t, p_t) = \frac{\exp(\alpha p_{jxt} + \xi_{jx} + \beta a_{jxt})}{\sum_k \exp(\alpha p_{kxt} + \xi_{kx} + \beta a_{kxt})}.$$

This estimates price sensitivity, promotional effects, and brand-size preference terms.

Step 2: inclusive values. For each size x , the inclusive value is

$$\omega_{xt} = \log \left\{ \sum_k \exp(\alpha p_{kxt} + \xi_{kx} + \beta a_{kxt}) \right\}.$$

The inclusive value summarizes the quality-adjusted price attractiveness of all brands available in a size. The authors estimate transition processes for the vector of inclusive values.

Step 3: dynamic quantity choice. The dynamic problem is then solved using the inclusive values as state variables. The household chooses the purchase size and consumption. The third stage estimates utility-from-consumption and storage-cost parameters by matching the observed sequence of purchase sizes.

5.4 Why the decomposition works

The decomposition works because, conditional on size, optimal consumption is independent of the brand purchased. With extreme value errors, the expected value of all brand choices within a size collapses to the inclusive value. Thus the dynamic problem can be written as a smaller quantity-choice problem rather than as a full brand-size-inventory problem.

6 Identification

6.1 Static parameters

Price sensitivity and promotional effects are identified from variation in prices, feature advertising, displays, and brand choices conditional on purchase size. Household heterogeneity is incorporated through demographic interactions and household-brand fixed effects.

6.2 Dynamic parameters

The dynamic parameters are identified from the relationship between current purchase probabilities, current inclusive values, past purchase quantities, and duration since previous purchases. If purchase probability did not depend on past purchases, the model would suggest no meaningful inventory dynamics. In contrast, if past purchase behavior predicts current purchase behavior, that supports an inventory model.

6.3 Storage costs

Storage costs affect purchase timing and the ability to exploit sales. Given similar total purchases, a household that buys more frequently in smaller amounts is inferred to have higher storage costs than a household that buys less frequently and stockpiles more.

7 Main estimation results

7.1 Brand-choice estimates

The first-stage estimates show that price matters, but the price coefficient changes substantially when feature and display variables are included. In specifications without promotional controls, the estimated price sensitivity is much larger. This implies that promotional activity is correlated with price reductions and must be controlled for separately.

7.2 Heterogeneity

Household characteristics matter. Larger families, nonwhite households, and suburban shoppers are estimated to be more price sensitive. The paper also allows for household-brand fixed effects, capturing persistent brand preferences.

7.3 Price process

The inclusive values are persistent, especially within the same size. Allowing different price processes for household types changes transition parameters, but has little impact on the main economic conclusions.

7.4 Dynamic storage and utility estimates

The third-stage estimates imply heterogeneous storage costs across household types. Suburban households, who are more likely to live in larger houses, have lower storage costs. The estimated median inventory is about 27 to 33 ounces depending on type, and mean weekly consumption is roughly 22 to 36 ounces across types.

7.5 Model fit

The model fits the hazard rate of purchases better than a static model. The empirical hazard rate increases after the previous purchase, peaks around week 4, and then declines. The model captures this broad interpurchase-duration pattern.

8 Elasticities and policy implications

8.1 Long-run elasticities

The long-run elasticities are simulated by changing the entire path of prices, re-solving the dynamic problem, and comparing simulated choice probabilities. This captures the response to permanent price changes, not just temporary sales.

8.2 Own-price elasticities

The dynamic model gives long-run own-price elasticities between roughly -2.5 and -4.5 for the main 128-ounce products. Static models overstate these own-price elasticities by about 30 percent because temporary sales induce purchase timing and stockpiling effects.

8.3 Cross-price elasticities

Cross-price elasticities to other brands are larger in the dynamic model than in the static model. A permanent increase in the price of one brand leads consumers to substitute across brands at all relevant inventory levels. A temporary price increase in a static model often mostly changes purchase timing, so it understates long-run cross-brand substitution.

8.4 No-purchase option

Static models greatly overstate substitution to no purchase. A temporary price increase may lead a household with inventory to delay purchase. A static model interprets this delay as substitution to the outside option. In the long run, however, a permanent price increase is more likely to shift consumers toward other brands or sizes rather than to eliminate detergent consumption.

8.5 Implications for margins

Demand elasticities are often used with first-order conditions to infer price-cost margins. For single-product firms, overstating own-price elasticities by 30 percent implies underestimating margins by roughly the same proportion. For multiproduct firms, the bias can be larger because static models also understate within-firm and cross-product substitution.

8.6 Implications for merger simulation

Static demand estimates tend to underpredict merger effects because they make products look less substitutable in the long run and make the outside option look too attractive. This can favor merger approval by understating the price effects of mergers and by defining markets too broadly.

9 Strengths and limitations

9.1 Strengths

- The paper connects a clear empirical phenomenon—sales and stockpiling—to structural demand estimation and antitrust policy.
- The likelihood decomposition makes a dynamic model feasible with household panel data and many differentiated products.
- The paper estimates long-run elasticities directly rather than relying on high-frequency static responses to temporary sales.
- The use of household-level store-visit data allows the authors to observe no-purchase decisions, not just purchases.

9.2 Limitations

- The model assumes that only total inventory matters, not brand-specific inventory.
- The price process is summarized by inclusive values and is assumed exogenous.
- Persistent unobserved heterogeneity in brand preferences is limited by the likelihood decomposition.
- The analysis focuses on the demand side and does not model firms' optimal sale-setting or retailer/manufacturer interactions.

9.3 Useful interpretation

The paper should be read as a warning about interpreting sales-based demand variation. A large purchase response to a temporary sale is not necessarily a large consumption response. For storable goods, high-frequency price variation identifies a mixture of brand substitution, purchase acceleration, inventory accumulation, and consumption changes.

10 Figures and tables

10.1 Table I. Summary statistics of household-level data and Table II. Brand volume shares and fraction sold on sale

TABLE I
SUMMARY STATISTICS OF HOUSEHOLD-LEVEL DATA^a

	Mean	Median	Std	Min	Max
Demographics					
Income (000's)	35.4	30.0	21.2	<10	>75
Size of household	2.6	2.0	1.4	1	6
Live in suburb	0.53	—	—	0	1
Purchase of laundry detergents					
Price (\$)	4.38	3.89	2.17	0.91	16.59
Size (oz.)	80.8	64	37.8	32	256
Quantity	1.07	1	0.29	1.00	4
Duration (days)	43.7	28	47.3	1	300
Number of brands bought over the 2 years	4.1	3	2.7	1	15
Brand HHI	0.53	0.47	0.28	0.10	1.00
Store visits					
Number of stores visited over the 2 years	2.38	2	1.02	1	5
Store HHI	0.77	0.82	0.21	0.27	1.00

^aFor Demographics, Store visits, Number of brands, and Brand HHI, an observation is a household. For all other statistics, an observation is a purchase instance. Brand HHI is the sum of the square of the volume share of the brands bought by each household. Similarly, Store HHI is the sum of the square of the expenditure share spent in each store by each household.

Table I. Summary statistics of household-level data.

TABLE II
BRAND VOLUME SHARES AND FRACTION SOLD ON SALE^a

Liquid					Powder				
Brand	Firm	Share	Cumulative	% on Sale	Brand	Firm	Share	Cumulative	% on Sale
1 Tide	P & G	21.4	21	32.5	Tide	P & G	40	40	25.1
2 All	Unilever	15	36	47.4	Cheer	P & G	14.7	55	9.2
3 Wisk	Unilever	11.5	48	50.2	A & H	C & D	10.5	65	28
4 Solo	P & G	10.1	58	7.2	Dutch	Dial	5.3	70	37.6
5 Purex	Dial	9	67	63.1	Wisk	Unilever	3.7	74	41.2
6 Cheer	P & G	4.6	72	23.6	Oxydol	P & G	3.6	78	59.3
7 A & H	C & D	4.5	76	21.5	Surf	Unilever	3.2	81	11.6
8 Ajax	Colgate	4.4	80	59.4	All	Unilever	2.3	83	
9 Yes	Dow Chemical	4.1	85	33.1	Dreft	P & G	2.2	86	15.2
10 Surf	Unilever	4	89	42.5	Gain	P & G	1.9	87	16.7
11 Era	P & G	3.7	92	40.5	Bold	P & G	1.6	89	1.1
12 Generic	—	0.9	93	0.6	Generic	—	0.7	90	16.6
13 Other	—	0.2	93	0.9	Other	—	0.6	90	19.9

^aColumns labeled Share are shares of volume (of liquid or powder) sold in our sample, columns labeled Cumulative are the cumulative shares, and columns labeled % on Sale are the percent of the volume, for that brand, sold on sale. A sale is defined as any price at least 5 percent below the modal price, for each UPC in each store. A & H = Arm & Hammer; P & G = Procter and Gamble; C & D = Church and Dwight.

Table II. Brand volume shares and fraction sold on sale.

Interpretation. Table I shows that households buy laundry detergent infrequently, with median duration of 28 days between purchases and median purchase size of 64 ounces. Table II shows a concentrated brand structure but substantial variation in sale intensity across brands, which creates useful price variation for estimating demand.

10.2 Table III. Quantity discounts and sales and Figure 1. Hazard rate of purchases

TABLE III
QUANTITY DISCOUNTS AND SALES^a

	Quantity Discount (%)	Quantity Sold on Sale (%)	Weeks on Sale (%)	Average Sale Discount (%)	Quantity Share (%)
Liquid					
32 oz.	—	2.6	2.0	11.0	1.6
64 oz.	18.1	27.6	11.5	15.7	30.9
96 oz.	22.5	16.3	7.6	14.4	7.8
128 oz.	22.8	45.6	16.6	18.1	54.7
256 oz.	29.0	20.0	9.3	11.8	1.6
Powder					
32 oz.	—	16.0	7.7	14.5	10.1
64 oz.	10.0	30.5	16.6	12.9	20.3
96 oz.	14.9	17.1	11.5	11.7	14.4
128 oz.	30.0	36.1	20.8	15.1	23.2
256 oz.	48.7	12.9	10.8	10.3	17.3

^a All cells are based on data from all brands in all stores. The column labeled Discount presents percent quantity discount (per unit) for the larger sizes, after correcting for differences across stores and brands (see text for details). The columns labeled Quantity Sold on Sale, Weeks on Sale, and Average Sale Discount present, respectively, the percent quantity sold on sale, percent of weeks a sale was offered, and average percent discount during a sale, for each size. A sale is defined as any price at least 5 percent below the modal. The column labeled Quantity Share is the share of the total quantity (measured in ounces) sold in each size.

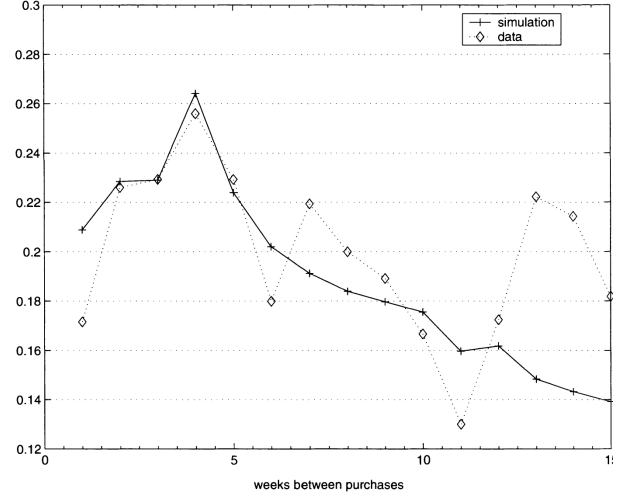


FIGURE 1.—Hazard rate of purchases.

Table III. Quantity discounts and sales.

Figure 1. Hazard rate of purchases.

Interpretation. Table III documents both nonlinear pricing and sales by size. Larger containers have lower unit prices, so storage costs are central for deciding whether to stockpile. Figure 1 shows that the dynamic model matches the hump-shaped purchase hazard reasonably well: purchase probability rises after the last purchase, peaks, and then gradually falls.

10.3 Table IV. First step: brand choice conditional on size and Table V. Second step: estimates of the price process

TABLE IV
FIRST STEP: BRAND CHOICE CONDITIONAL ON SIZE^a

	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
Price	-0.51 (0.022)	-1.06 (0.038)	-0.49 (0.043)	-0.26 (0.050)	-0.27 (0.052)	-0.38 (0.055)	-0.38 (0.056)	-0.57 (0.083)	-1.41 (0.092)	-0.75 (0.098)
*Suburban dummy				-0.33 (0.055)	-0.30 (0.061)	-0.34 (0.055)	-0.33 (0.056)	-0.25 (0.113)	-0.45 (0.127)	-0.19 (0.127)
*Nonwhite dummy				-0.34 (0.075)	-0.39 (0.083)	-0.38 (0.076)	-0.33 (0.076)	-0.34 (0.152)	-0.26 (0.166)	-0.26 (0.168)
Large family				-0.23 (0.080)	-0.13 (0.107)	-0.21 (0.080)	-0.22 (0.082)	-0.46 (0.181)	-0.38 (0.192)	-0.43 (0.195)
Feature		1.06 (0.095)	1.05 (0.096)	1.08 (0.097)	0.92 (0.099)	0.93 (0.100)	1.08 (0.123)	1.05 (0.126)	1.05 (0.126)	1.52 (0.193)
Display		1.19 (0.069)	1.17 (0.070)	1.20 (0.071)	1.14 (0.071)	1.15 (0.072)	1.55 (0.093)	1.52 (0.093)	1.52 (0.093)	1.52 (0.093)
Brand dummy variable		✓	✓	✓	✓	✓	✓	✓	✓	✓
*Demographics										
*Size										
Brand-size dummy variable										
Brand-HH dummy variable										
*Size										

^a Estimates of a conditional logit model. An observation is a purchase instance by a household. Options include only products of the same size as the product actually purchased. Asymptotic standard errors are shown in parentheses.

Table IV. First step: brand choice conditional on size.

quantity weighted price. For each household, all the different prices (and promotional activities) of all the products offered in each size are combined into

TABLE V
SECOND STEP: ESTIMATES OF THE PRICE PROCESS^a

	Same Process for All Types				Different Process for Each Type			
	ω_{2t}	ω_{4t}	ω_{2t}	ω_{4t}	ω_{2t}	ω_{4t}	ω_{2t}	ω_{4t}
$\omega_{1,t-1}$	0.003 (0.012)	-0.014 (0.011)	0.005 (0.014)	0.014 (0.014)	-0.023 (0.017)	-0.005 (0.014)	-0.019 (0.019)	0.007 (0.015)
$\omega_{2,t-1}$	0.413 (0.007)	0.033 (0.010)	0.295 (0.008)	0.025 (0.007)	0.575 (0.013)	-0.003 (0.010)	0.520 (0.016)	0.011 (0.013)
$\omega_{3,t-1}$	0.003 (0.007)	-0.034 (0.007)	0.041 (0.009)	-0.006 (0.009)	0.027 (0.020)	-0.072 (0.016)	0.051 (0.025)	-0.018 (0.020)
$\omega_{4,t-1}$	0.029 (0.008)	0.249 (0.008)	0.026 (0.008)	0.236 (0.017)	-0.018 (0.020)	0.336 (0.016)	-0.018 (0.021)	0.274 (0.017)
$\sum_{\tau=2}^5 \omega_{1,t-\tau}$			-0.003 (0.005)	-0.012 (0.004)			-0.008 (0.006)	-0.003 (0.005)
$\sum_{\tau=2}^5 \omega_{2,t-\tau}$			0.089 (0.003)	0.006 (0.002)			0.073 (0.005)	-0.004 (0.004)
$\sum_{\tau=2}^5 \omega_{3,t-\tau}$			-0.008 (0.003)	-0.009 (0.003)			-0.004 (0.008)	-0.016 (0.006)
$\sum_{\tau=2}^5 \omega_{4,t-\tau}$			-0.013 (0.003)	0.018 (0.003)			-0.008 (0.008)	0.056 (0.006)

Table V. Second step: estimates of the price process.

Interpretation. Table IV shows that the brand-choice price coefficient falls sharply once feature and display controls are included, highlighting the importance of controlling separately for nonprice

promotions. Table V summarizes the inclusive-value transition process; own-size lagged inclusive values are the main predictors of future inclusive values.

10.4 Table VI. Third step: estimates of dynamic parameters and Table VII. Long-run own- and cross-price elasticities

TABLE VI
THIRD STEP: ESTIMATES OF DYNAMIC PARAMETERS^a

Household Type:	1	2	3	4	5	6
	Urban Market			Suburban Market		
Household Size:	1-2	3-4	5+	1-2	3-4	5+
Cost of inventory						
Linear	9.24 (0.01)	6.49 (0.02)	21.96 (0.09)	4.24 (0.01)	4.13 (0.17)	11.75 (5.3)
Quadratic	-3.82 (29.8)	1.80 (1.77)	-35.86 (0.19)	-8.20 (0.03)	-6.14 (1.69)	-0.73 (1.53)
Utility from consumption	1.31 (0.02)	0.75 (0.09)	0.51 (0.21)	0.08 (0.03)	0.92 (0.18)	3.80 (0.38)
Log likelihood	365.6	926.8	1,530.1	1,037.1	543.6	1,086.1

^aAsymptotic standard errors are shown in parentheses. Also included are size fixed effects, which are allowed to vary by household type.

Table VI. Third step: estimates of dynamic parameters.

TABLE VII
LONG-RUN OWN- AND CROSS-PRICE ELASTICITIES^a

Brand	Size (oz.)	All ^b	Wisk	Surf	Cheer	Tide	Private Label
All ^b	32	0.418	0.129	0.041	0.053	0.131	0.000
	64	0.482	0.093	0.052	0.033	0.085	0.006
	96	0.725	0.092	0.036	0.035	0.100	0.002
	128	-2.536	0.154	0.088	0.059	0.115	0.007
Wisk	32	0.088	0.702	0.046	0.012	0.143	0.006
	64	0.078	0.620	0.045	0.014	0.116	0.004
	96	0.066	0.725	0.051	0.022	0.135	0.009
	128	0.126	-2.916	0.083	0.026	0.147	0.005
Surf	32	0.047	0.061	0.977	0.024	0.369	0.003
	64	0.146	0.086	0.905	0.023	0.158	0.005
	96	0.160	0.101	0.915	0.016	0.214	0.001
	128	0.202	0.149	-3.447	0.039	0.229	0.008
Cheer	64	0.168	0.049	0.027	0.831	0.293	0.001
	96	0.167	0.015	0.008	0.982	0.470	0.001
	128	0.250	0.090	0.058	-4.341	0.456	0.003
	32	0.071	0.085	0.050	0.022	1.007	0.002
Tide	64	0.048	0.055	0.024	0.025	0.924	0.001
	96	0.045	0.063	0.016	0.026	1.086	0.001
	128	0.072	0.093	0.039	0.045	-2.683	0.001
	64	0.066	0.070	0.027	0.021	0.150	0.002
Solo	96	0.219	0.032	0.023	0.033	0.075	0.000
	128	0.127	0.125	0.060	0.043	0.302	0.001
	32	0.035	0.155	0.039	0.022	0.425	0.000
	64	0.030	0.103	0.039	0.018	0.304	0.008
Era	96	0.035	0.168	0.033	0.027	0.352	0.001
	128	0.054	0.192	0.061	0.029	0.513	0.014
	64	0.123	0.119	0.066	0.039	0.081	0.248
	128	0.174	0.266	0.100	0.019	0.072	-2.682
Private label	No purchase	0.007	0.002	0.004	0.000	0.013	0.000

^aCell entries i and j , where i indexes row and j indexes column, give the percent change in market share of brand i with a 1 percent change in the price of j . All columns are for a 128 oz. product, the most popular size. The results are based on Tables IV-VI.

^bNote that "All" is the name of a detergent produced by Unilever.

Table VII. Long-run own- and cross-price elasticities.

Interpretation. Table VI reports dynamic parameters by household type. The implied storage costs vary across markets and household sizes. Table VII reports long-run elasticities from the dynamic model; own-price elasticities are large, but cross-price elasticities across sizes and brands are also economically meaningful.

10.5 Table VIII. Ratios of static elasticities to dynamic long-run elasticities

TABLE VIII
AVERAGE RATIOS OF ELASTICITIES COMPUTED FROM A STATIC MODEL TO LONG-RUN ELASTICITIES COMPUTED FROM THE DYNAMIC MODEL^a

Brand	Size (oz.)	64 oz.						128 oz.					
		All ^b	Wisk	Surf	Cheer	Tide	Private Label	All ^b	Wisk	Surf	Cheer	Tide	Private Label
All ^b	64	1.03	0.13	0.14	0.12	0.13	0.15	0.14	0.17	0.17	0.18	0.21	0.34
	128	0.17	0.24	0.26	0.20	0.28	0.35	1.23	0.09	0.11	0.09	0.15	0.22
Wisk	64	0.14	1.20	0.13	0.17	0.12	0.13	0.16	0.22	0.14	0.22	0.25	0.20
	128	0.25	0.27	0.23	0.31	0.26	0.28	0.08	1.42	0.08	0.13	0.18	0.11
Surf	64	0.14	0.13	0.93	0.16	0.13	0.14	0.18	0.18	0.12	0.18	0.22	0.28
	128	0.25	0.22	0.18	0.27	0.25	0.18	0.12	0.11	1.20	0.08	0.15	0.14
Cheer	64	0.12	0.17	0.16	0.84	0.09	0.13	0.14	0.24	0.16	0.14	0.22	0.24
	128	0.25	0.26	0.26	0.12	0.23	0.22	0.09	0.12	0.06	0.89	0.15	0.07
Tide	64	0.16	0.17	0.13	0.13	1.26	0.15	0.22	0.28	0.16	0.26	0.22	0.37
	128	0.25	0.31	0.22	0.24	0.22	0.31	0.11	0.16	0.08	0.13	1.44	0.31
Solo	64	0.15	0.12	0.15	0.14	0.12	0.14	0.17	0.15	0.15	0.30	0.30	0.28
	128	0.23	0.20	0.24	0.21	0.21	0.25	0.07	0.07	0.06	0.16	0.17	0.21
Era	64	0.21	0.12	0.13	0.13	0.10	0.19	0.43	0.17	0.15	0.22	0.19	0.35
	128	0.31	0.22	0.24	0.25	0.17	0.38	0.19	0.08	0.09	0.11	0.10	0.22
Private label	64	0.19	0.15	0.14	0.17	0.17	1.02	0.32	0.22	0.15	0.26	0.31	0.25
	128	0.29	0.28	0.34	0.30	0.39	0.29	0.16	0.12	0.13	0.10	0.27	1.29
No purchase		2.12	1.13	1.15	1.40	1.27	2.39	1.80	7.60	2.26	14.11	2.38	10.86

^aCell entries i and j , where i indexes row and j indexes column, give the ratio of the (short-run) elasticities computed from a static model divided by the long-run elasticities computed from the dynamic model. The elasticities for both models are the percent change in market share of brand i with a 1 percent change in the price of j . The static model is identical to the model estimated in the first step, except that brands of all sizes are included as well as a no-purchase decision, not just products of the same size as the chosen option. The results from the dynamic model are based on the results presented in Tables IV-VI.

^bNote that "All" is the name of a detergent produced by Unilever.

Table VIII. Ratios of static elasticities to dynamic long-run elasticities.

Interpretation. Table VIII is the key policy table. Static own-price elasticities are generally larger than dynamic own-price elasticities, but static cross-price elasticities to other products are much smaller than dynamic cross-price elasticities. The static model also overstates substitution to no purchase, sometimes dramatically.

11 Takeaways for future paper reading

1. For storable goods, temporary price cuts identify short-run purchase timing and inventory responses, not just long-run demand.
2. Dynamic models can change both own-price and cross-price elasticities in policy-relevant ways.
3. Inventory behavior can make the outside option look too attractive in static models.
4. Conditional brand choice and inclusive values can be used as a computational shortcut when the full state space is too large.
5. The economic interpretation of sales depends on whether the researcher cares about short-run volume responses or long-run consumption and substitution responses.